



**MRIM Installation and Operations Manual**

**Railway Administration and Information Logical System**

*RAILS for Model Railroads*

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# Chapter 1

## Model Railroad Information Manager

### 1.1 Introduction

Railway Administration and Information Logical System (RAILS) is a software model and implementation of an automated system to assist the model railroader achieve realism in the operation of a model railroad. The RAILS system design describes the system's architecture, components, and interfaces and can be found at [RAILS System Design and implementation](#). There are four user interface Single Page Applications (SPAs) that provide different aspects of RAILS they are:

- Rollingstock RFID Manager (RSRM) provides a user with a web application to show, when an RFID tag is read, the road name and number of the rolling stock associated with that tag.
- Model Railroad Inventory Manager (MRIM) the focus of this manual, allows the user to create, update and delete model railroad assets, such as:
  - manage lists of items found in model railroads:
    - ✱ Rolling Stock
    - ✱ DCC decoders
    - ✱ AAR Codes
    - ✱ Images
    - ✱ Companies
    - ✱ Structures
  - creates PDF reports; and
    - ✱ AAR Codes
    - ✱ Companies
    - ✱ Images
    - ✱ RFID
    - ✱ Rolling Stock sortable by Road Name and Road Number or AAR Code or Status
  - exports and imports inventories in CSV files.
- Model Projects and Purchase Manager (MPPM) allows the user to enter information about their projects and purchases; and
- Model Railroad Layout Manager (MRLM) allows the user to enter information about their layout and control elements of it.

## 1.2 Components

The implementation of MRIM consists of the following microservices components:

- Model Railroad File Manager (MRFM) provides the user the ability to upload image files;
- images is a repository for the images of rollingstock and structures;
- Railroad Inventory Data Services (RIDS) provides Representational State Transfer (REST) access to railroad inventory collections including rollingstock (RS);
- MongoDB a NoSQL database program that stores data records as documents which are gathered in collections. A database stores one or more collections of documents;
- Model Railroad (MR) Data is the document repository, used by MongoDB, to store complete collections of items such as RS, industries (producers and consumers), track elements, turnouts, projects, purchases, etc. in support of RAILS; and
- MRIM is the SPA.

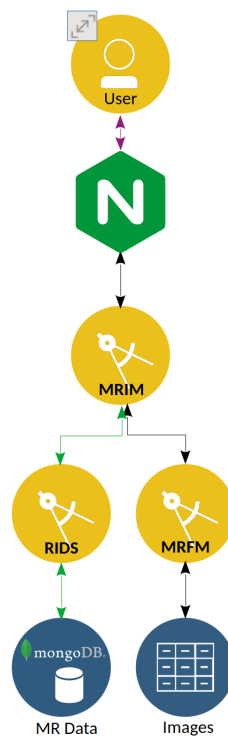


Figure 1.1: Microservices Components

## 1.3 System Components

Three system components that make up this subsystem:

- Network is a Transmission Control Protocol/Internet Protocol (TCP/IP) communication medium that connects the client (one or more) to the host running the MRIM docker containers;
- Server is a computer that runs MRIM microservices components; and
- Client<sup>1</sup> is a computer that runs a web browser to display the MRIM SPA.

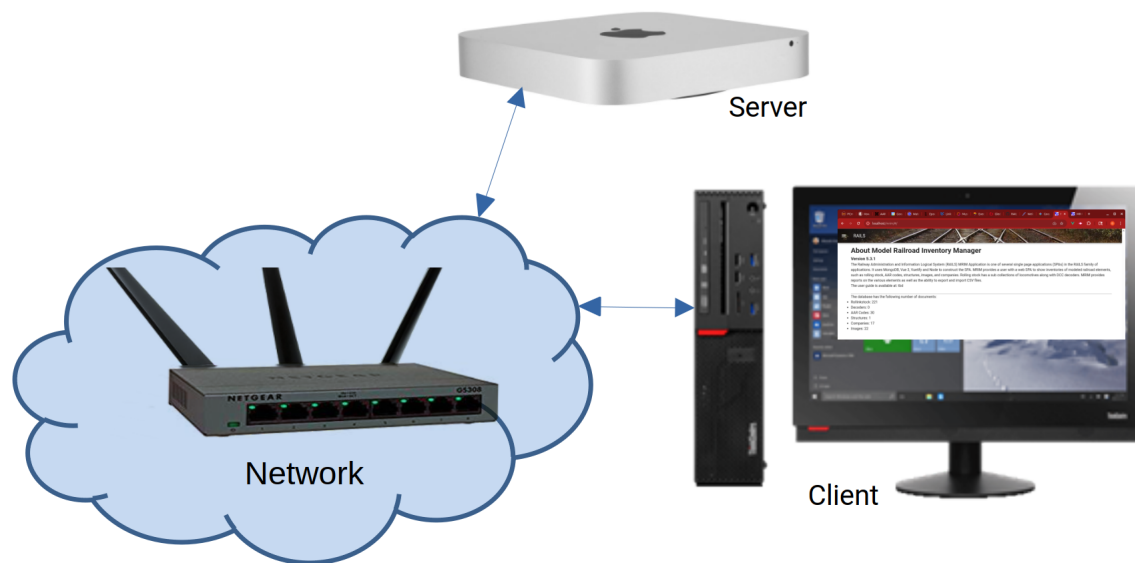


Figure 1.2: System Components

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<sup>1</sup>A web browser may be used on the Server and client machine is omitted

## Chapter 2

# Docker Set Up

### 2.1 Docker Installation

The RAILS SPAs are implemented as Docker containers. The Docker containers are built and pushed to Docker Hub. The Docker containers are then pulled from Docker Hub and run on the host machine. The host machine must have Docker installed. Docker is a platform for developing, shipping, and running applications in containers. Docker can be installed on Windows, macOS, and Linux. The installation instructions for Docker can be found at <https://docs.docker.com/get-docker/>.

### 2.2 Docker Compose Installation

Docker Compose is a tool for defining and running multi-container Docker applications. Docker Compose uses a yet another markup language (YAML) file to configure the application's services. The installation instructions<sup>1</sup> for Docker Compose can be found at <https://docs.docker.com/compose/install/>.

### 2.3 Broker Configuration

The RAILS SPAs use the Mosquitto broker to communicate with each other. The Mosquitto broker is an open-source message broker that implements the Message Queuing Telemetry Transport (MQTT) protocol. The Mosquitto broker must be configured to allow the RAILS SPAs to communicate with each other. The Mosquitto broker configuration file must be edited to allow the RAILS SPAs to communicate with each other. The configuration file is found in the virtual storage whose volume name is "mosquitto". To establish and modify the configuration file the following steps are taken:

1. Create a folder/directory "RAILS" and open a terminal from that folder/directory (all docker commands should be run in that terminal).
2. Create a Docker volume for the Mosquitto broker.

---

<sup>1</sup>Note that for Microsoft Windows the installation of the Graphic User Interface (GUI) Docker Desktop automatically installs Docker Compose.

```
docker volume create --name mosquitto
```

3. Run a Docker container for the first time the Mosquitto broker, which will create the initial configuration file.

```
docker run -it --name myMqttBrkr -p 1883:1883 -p 9001:9001 --rm  
-v mosquitto:/mosquitto -d eclipse-mosquitto
```

4. Stop the broker container.

```
docker stop myMqttBrkr
```

5. Locate the configuration file in the "mosquitto" volume.

```
docker inspect mosquitto
```

6. Edit the configuration file modifying or adding the following lines:

```
listener 1883  
allow_anonymous true  
socket_domain ipv4
```

## 2.4 Create the RAILS Docker Environment

The **RAILS SPAs** are implemented as Docker containers that require an environment to run in. To create that environment, the following steps are taken:

1. Create a Docker volume for the Rails database.

```
docker volume create --name myRailsDb
```

2. Create a Docker volume for the Rails images.

```
docker volume create --name myRailsImages
```

3. Create a Docker network for the Rails containers to communicate over.

```
docker network create myRailsNet
```

## 2.5 Modify the Model Railroad Name

To change the name of the model railroad displayed in the SPAs from "KJ&C RR" to a name of choice edit the docker-compose.yaml file. Find the text at line #67 for the SPA MPPM:

```
- MRNAME=KJ&C RR
```

Edit the string "KJ&C RR" to the name of choice. Then repeat for each of the other SPAs as follows:

- at line #104 for the SPA MRIM
- at line #152 for the SPA RSRM
- at line #197 for the SPA MRLM

## 2.6 Run the RAILS Docker Containers

The Docker containers are pulled from Docker Hub and run on the host machine. To run all of the RAILS Docker containers retrieve the YAML "docker-compose.yaml" file from the RAILS GitHub repository found at [my GitHub repository](#). Additionally retrieve the "nginx.conf" file from RAILS GitHub repository. Be sure that both files are put into the "RAILS" folder/directory. To fetch the Docker images, create the containers, and start the containers running execute the following command in the terminal opened in the "RAILS" folder/directory:

```
docker compose up -d
```

This command will pull the Docker images from Docker Hub and run the containers in detached mode. The RAILS SPAs will be running in the background, and the terminal will return to the command prompt. The RAILS SPAs will be running on the host machine. Point a browser, on the host machine, to the following URLs to access the RAILS SPAs:

- MRIM SPA - <http://localhost/mrim> (or [http://<MACHINE\\_IP>/mrim/](http://<MACHINE_IP>/mrim/))
- RSRM SPA - <http://localhost/rsrm> (or [http://<MACHINE\\_IP>/rsrm/](http://<MACHINE_IP>/rsrm/))
- MPPM SPA - <http://localhost/mppm> (or [http://<MACHINE\\_IP>/mppm/](http://<MACHINE_IP>/mppm/))
- MRLM SPA - <http://localhost/mrlm> (or [http://<MACHINE\\_IP>/mrlm/](http://<MACHINE_IP>/mrlm/))

Where <MACHINE\_IP> is the IP address of the host machine running the Docker containers. The RAILS SPAs can be accessed from any browser on any Personal Computer (PC) in the network.

## 2.7 Stop and Remove the RAILS Docker Containers

To stop and remove the RAILS Docker containers, the following command is used:

```
docker compose down
```

## Chapter 3

# MRIM Operations

RSRM simplifies identifying rollingstock by associating Radio Frequency Identification (RFID) tags with specific road names and numbers. When a connected reader reads an RFID tag, RSRM searches its database for a matching entry. If a match is found, the application displays the corresponding road name and number of the rollingstock. If no match is found, the application provides input fields for the user to manually enter this information, creating a new association in the database.

### 3.1 About Page

The intial page when the application is opened in a web browser provides:

- A version number of the application;
- A concise overview of the application's purpose and technology stack;
- The current database statistics indicating:
  - The number of rollingstock entries in the database and how many have RFID tags; and
  - The number of micro-controllers in the database and how many are RFID readers.

Figure 3.1 shows the "About" page of the RAILS RS RFID Manager application.

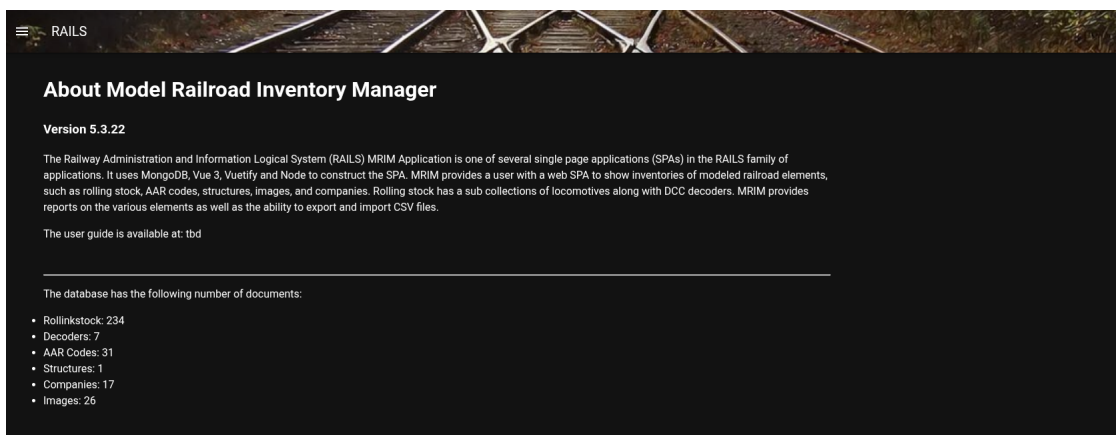


Figure 3.1: About Page

The hamburger menu icon ☰, to the left of "RAILS" represents a hidden navigation menu that can be toggled open and closed. When the icon is clicked or tapped on the menu for navigation to different pages of this application is revealed. This menu contains the links: "Reader," "Admin," "Rollingstock," "AAR Codes," "About" (this page). Figure 3.2 shows the "About" page with the navigation menu opened.

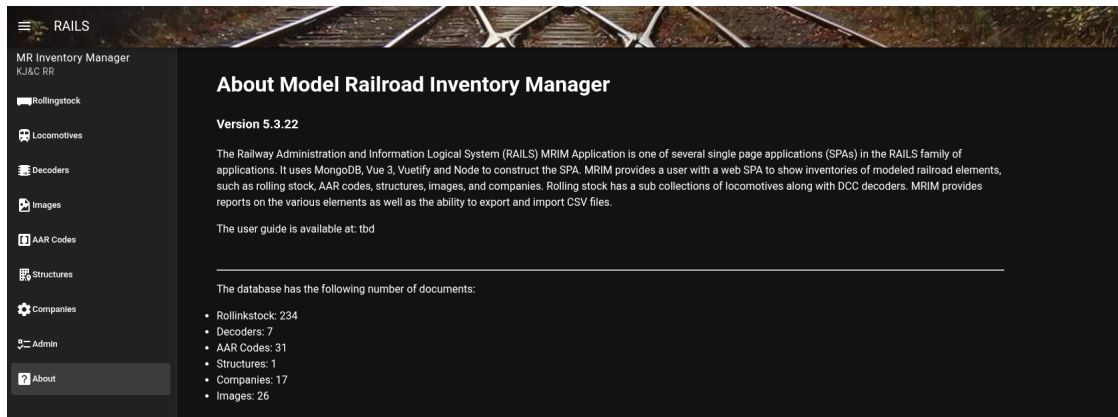


Figure 3.2: Sidemenu

## 3.2 Rolling Stock Management

The Rollingstock view provides a complete inventory of all rolling stock registered in the system. Selecting "Rollingstock" from the navigation drawer displays the main tabular view containing all registered pieces, including key attributes such as Road Name, Road Number, AAR Code, Description, Color, and available Action buttons. Appendix A Table A.5 defines all of the rollingstock attributes. Figure 3.3 displays the main Rollingstock table page.

| Road Name | Road Number | AAR | Description                | Color | Actions                                                                                                                                                                                                                                                           |
|-----------|-------------|-----|----------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ACFX      | 44763       | LO  | 2 Bay Covered Hopper       | Grey  |                                                                                         |
| ACFX      | 44867       | LO  | 2 Bay Covered Hopper       | Grey  |                                                                                         |
| ADCX      | 5425        | RM  | 36 ft Wooden Reefer        | White |                                                                                         |
| ALNX      | 396029      | LO  | Cylindrical Covered Hopper | Blue  |    |
| ALNX      | 396125      | LO  | Cylindrical Covered Hopper | Blue  |    |
| ALNX      | 396185      | LO  | Cylindrical Covered Hopper | Blue  |    |
| ALPX      | 628333      | LO  | Cylindrical Covered Hopper | Blue  |    |
| ALPX      | 628343      | LO  | Cylindrical Covered Hopper | Blue  |    |
| ALPX      | 628416      | LO  | Cylindrical Covered Hopper | Blue  |    |
| ALPX      | 628459      | LO  | Cylindrical Covered Hopper | Blue  |    |

Figure 3.3: Rollingstock Page

### 3.2.1 Page Quantity Selection

The table displays items in paginated format. Users can control the number of rolling stock items shown per page using the dropdown selector located at the bottom right of the table. Available options include 10, 25, 50, 100, or All items. Figure 3.4 illustrates changing the items per page display count.

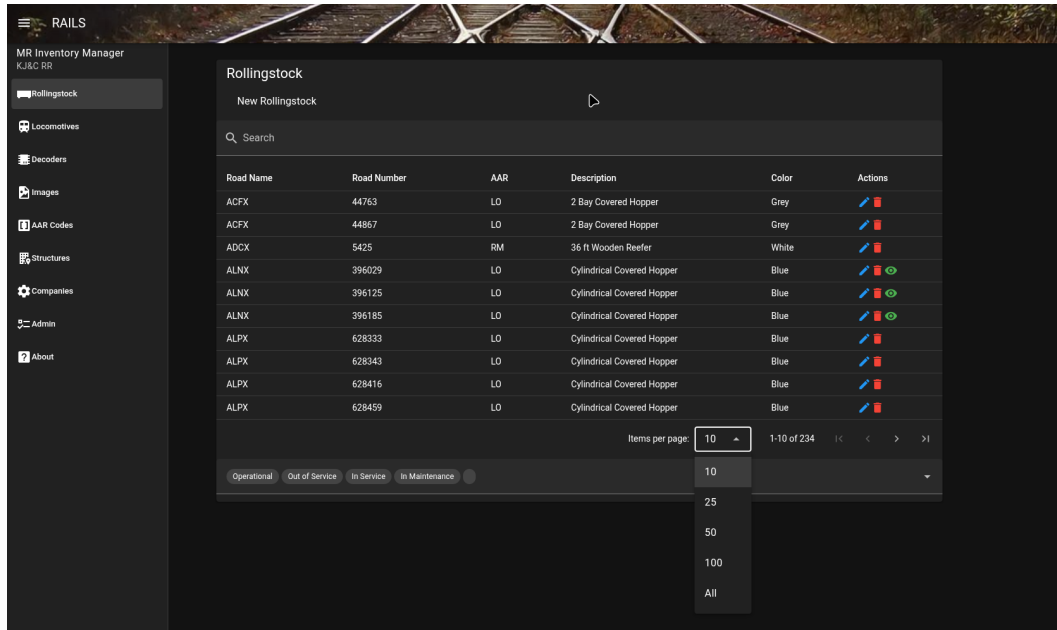


Figure 3.4: Items Per Page Selection

### 3.2.2 Status Filtering

Rolling stock entries can be filtered dynamically based on operational status. At the bottom of the view, status filter controls allow toggling visibility for items designated as Operational, Out of Service, In Service, or In Maintenance. Figure 3.5 shows the operational status filtering interface open.

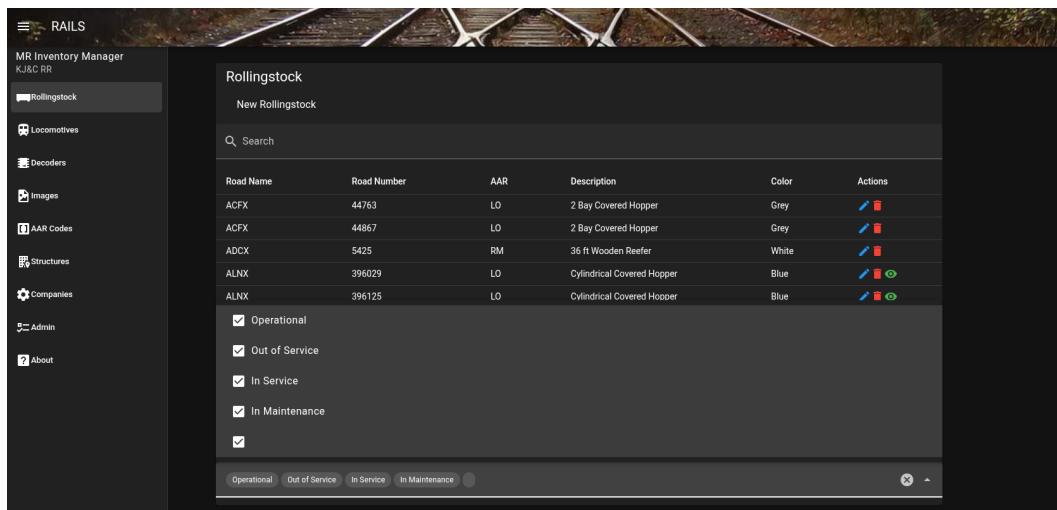


Figure 3.5: Status Filter

### 3.2.3 Search Functionality

A search bar located above the main table enables real-time filtering across rolling stock attributes. Typing search terms such as a road name (e.g., "cnw") instantly filters the list to show matching records. Figure 3.6 shows filtered search results.

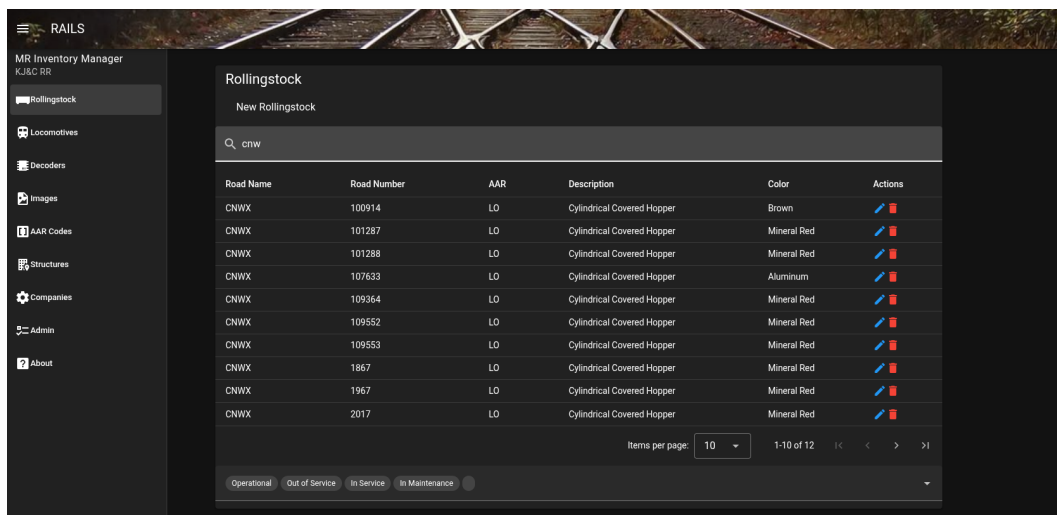


Figure 3.6: Search Rollingstock

### 3.2.4 Viewing Rollingstock Details

For entries configured with associated image files, an eye icon appears under the Actions column. Clicking this icon opens the Rollingstock Details dialog, displaying comprehensive prototype specifications, dimensions, status, and a photograph of the equipment. Figure 3.7 shows the detailed view modal for a selected car.

RAILS

### Rollingstock Details

Road Name: ALNX  
Road Number: 396029  
AAR Code: LO  
Color: Blue  
Description: Cylindrical Covered Hopper

Number Built: 0  
Builder: NSC  
Built Date:  
In Service Date: 1981-02-24  
Inside Length: 59  
Inside Height:  
Inside Width:  
L1 Weight: 61800  
Load Limit: 201200  
Load Types: Grain  
Capacity: 4550  
Home Location:  
Last Maintenance: 1975-06-28  
Number of Axles: 4  
Next Maintenance:  
Status: Operational  
Model Details  
RFID Tag: AAADCDDAE5  
RFID Location: 0  
Image ID: 45103wl.jpg  
Weight: 0  
Length: 0  
Notes:  
Close

| Description                | Color | Actions |
|----------------------------|-------|---------|
| Cylindrical Covered Hopper | Blue  |         |
| Cylindrical Covered Hopper | Blue  |         |
| 1 Wooden Reefer            | White |         |
| Cylindrical Covered Hopper | Blue  |         |
| Cylindrical Covered Hopper | Blue  |         |
| Cylindrical Covered Hopper | Blue  |         |
| Cylindrical Covered Hopper | Blue  |         |
| Cylindrical Covered Hopper | Blue  |         |
| Cylindrical Covered Hopper | Blue  |         |
| Cylindrical Covered Hopper | Blue  |         |

Items per page: 10 1-10 of 235

Figure 3.7: View Rollingstock Details

### 3.2.5 Adding New Rollingstock

Clicking the “New Rollingstock” button above the table opens a blank data entry form. Users can populate fields including road information, dimensions, load characteristics, model parameters, RFID tag associations, and linked image IDs. Figure 3.8 shows the entry form for creating a new rolling stock item.

The screenshot shows the RAILS application interface. On the left is a 'New Rollingstock' form with the following fields:

- Top row: Road Name, Road Number, AAR Code, Color
- Description (text area)
- Number Built (0), Builder, Built Date, In Service Date
- Inside Length, Inside Height, Inside Width, Lt Weight (0)
- Load Limit (0), Load Types, Capacity (0), Home Location
- Last Maintenance, Number of Axles (4), Status (dropdown)
- Model Details section:
  - RFID Tag, RFID Location (0), Image ID
  - Weight (0), Length (0)
  - Notes (text area)
- Buttons: Cancel, Save

On the right is a table of existing rollingstock records:

| AAR | Description                | Color | Actions                      |
|-----|----------------------------|-------|------------------------------|
| LO  | 2 Bay Covered Hopper       | Grey  | [Pencil] [Trash]             |
| LO  | 2 Bay Covered Hopper       | Grey  | [Pencil] [Trash]             |
| RM  | 36 ft Wooden Reefer        | White | [Pencil] [Trash]             |
| LO  | Cylindrical Covered Hopper | Blue  | [Pencil] [Trash] [Checkmark] |
| LO  | Cylindrical Covered Hopper | Blue  | [Pencil] [Trash] [Checkmark] |
| LO  | Cylindrical Covered Hopper | Blue  | [Pencil] [Trash] [Checkmark] |
| LO  | Cylindrical Covered Hopper | Blue  | [Pencil] [Trash]             |
| LO  | Cylindrical Covered Hopper | Blue  | [Pencil] [Trash]             |
| LO  | Cylindrical Covered Hopper | Blue  | [Pencil] [Trash]             |
| LO  | Cylindrical Covered Hopper | Blue  | [Pencil] [Trash]             |

Below the table is a pagination control: 'Items per page: 10', '1-10 of 235', and navigation arrows.

Figure 3.8: New Rollingstock Entry

### 3.2.6 Editing Rollingstock Records

To modify an existing entry, click the blue pencil icon in the Actions column for that row. This opens the details dialog in editable mode, populated with current record data for updating field values. Figure 3.9 illustrates editing an existing record.

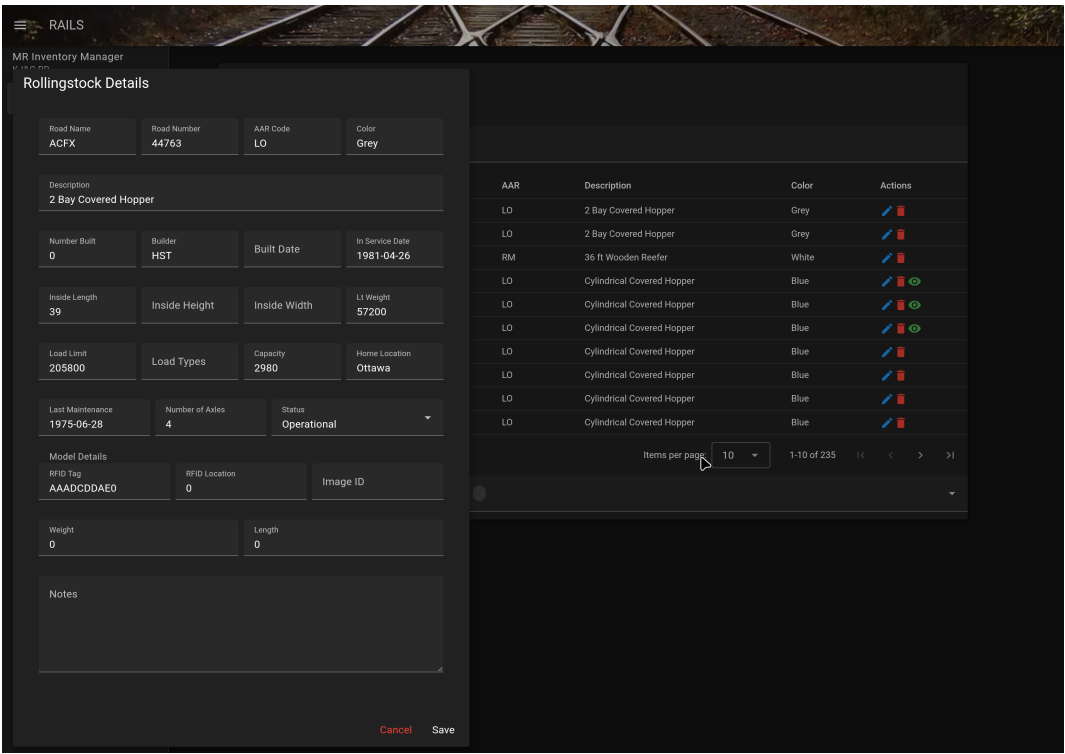


Figure 3.9: Edit Rollingstock

3.2.7 Deleting Rollingstock Records

To remove a record from the database, click the red trash bin icon in the Actions column. A confirmation dialog appears requesting confirmation before permanently removing the record. Figure 3.10 shows the deletion confirmation prompt.

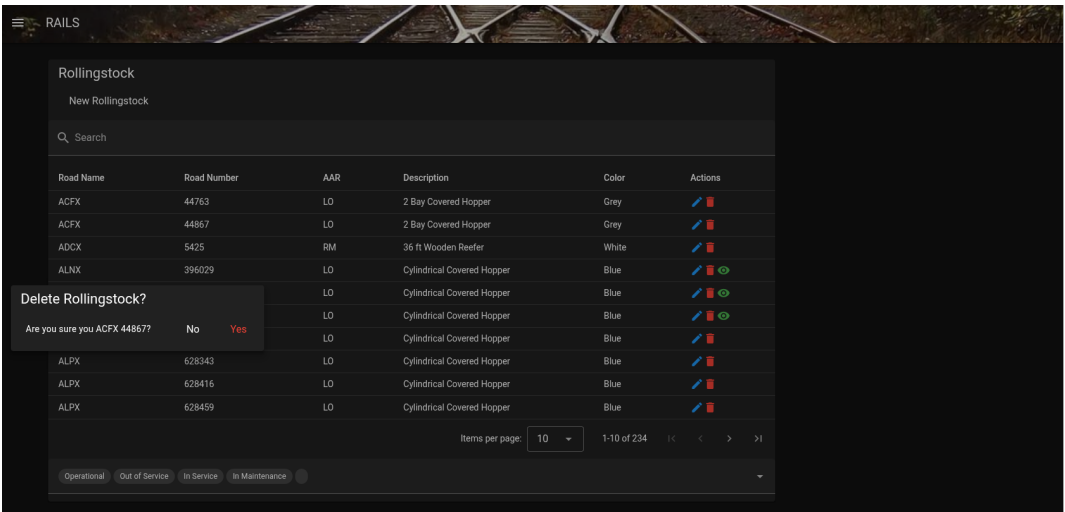


Figure 3.10: Delete Rollingstock Confirmation

### 3.3 Locomotives Inventory

Selecting "Locomotives" from the navigation drawer displays the Inventory of Locomotives view. This table presents a specialized subset of motive power entries from the inventory database, detailing primary attributes such as Road Name, Road Number, AAR designation code, Description, and Color scheme.

| Road Name | Road Number | AAR | Description                   | Color             |
|-----------|-------------|-----|-------------------------------|-------------------|
| ATSF      | 3275        | SE  | 2-8-2 Heavy Mikado Locomotive | Black             |
| CMC       | 2           | SE  | 0-6-0 Saddle Tank Locomotive  | Black             |
| CN        | 3015        | DE  | GE ET44 Locomotive            | Red/White/Camo    |
| CN        | 5347        | DE  | GP40-2                        | Black / Red       |
| CN        | 8625        | DE  | RS-2                          | Olive             |
| CN        | 9407        | DE  | FA-1                          | Olive / Yellow    |
| CN        | 9526        | DE  | GP40-2                        | Black / Red       |
| CP        | 5000        | DE  | GP30                          | Tuscan Red / Grey |
| K&J       | 177         | SE  | 0-4-0 Shifter                 | Black             |
| KJ&C      | 580         | DE  | F7A                           | Grey / Maroon     |

Figure 3.11: Inventory of Locomotives

The page incorporates interactive column sorting, as shown by the direction arrow next to the AAR header, allowing motive power records to be ordered by specific attributes. Standard table pagination controls located at the bottom right enable users to adjust the display limit per page and navigate through larger locomotive fleets.

### 3.4 Decoders Management

Selecting "Decoders" from the navigation drawer displays the main Decoders management view. This page displays a tabular listing of all Digital Command Control (DCC) decoders installed in inventory locomotives, summarizing information such as Road Name, Road Number, Manufacturer, Family, Model, active DCC Address, and available action controls. Figure 3.12 shows the main Decoders table page.

| Road Name | Road Number | Manufacturer                  | Family                            | Model                       | Address | Actions                                                                                                                                                                     |
|-----------|-------------|-------------------------------|-----------------------------------|-----------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ATSF      | 3275        | Broadway Limited              | Paragon 3                         | Rolling Thunder             | 3275    |   |
| CMC       | 2           | Bachmann Trains               | E-Z Command decoders              | 4 function decoder (36-550) | 2       |   |
| CN        | 3015        | Electronic Solutions Ulm GmbH | LokSound                          | 5                           | 3015    |   |
| CN        | 5347        | Digitrax                      | Economy Series 3 with FX3, silent | DH123 Four Function         | 5347    |   |
| KJ&C      | 580         | Digitrax                      | Economy Series 3 with FX3, silent | DH123 Four Function         | 580     |   |
| VIA       | 6922        | Electronic Solutions Ulm GmbH | LokSound                          | 5                           | 6922    |   |
| CN        | 8625        | Digitrax                      | Economy Series 3 with FX3, silent | DH123 Four Function         | 8625    |   |

Figure 3.12: Decoders Page

### 3.4.1 Adding a New Decoder

Clicking the “Add Decoder” button above the table opens a modal dialog for registering a new decoder entry. The form includes fields for Road Name, Road Number, Manufacturer, Family, Model, and DCC Address. Figure 3.13 shows the new decoder entry dialog.

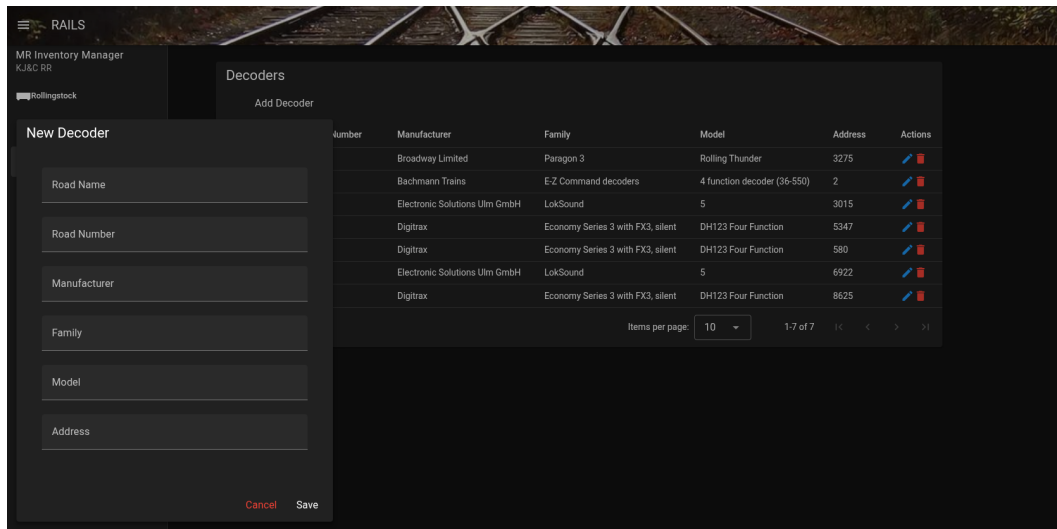


Figure 3.13: Add Decoder

### 3.4.2 Editing Decoder Records

To modify an existing decoder entry, click the blue pencil icon under the Actions column. This opens the decoder configuration form pre-populated with current settings, allowing updates to hardware details or DCC address assignments. Figure 3.14 displays the edit decoder interface.

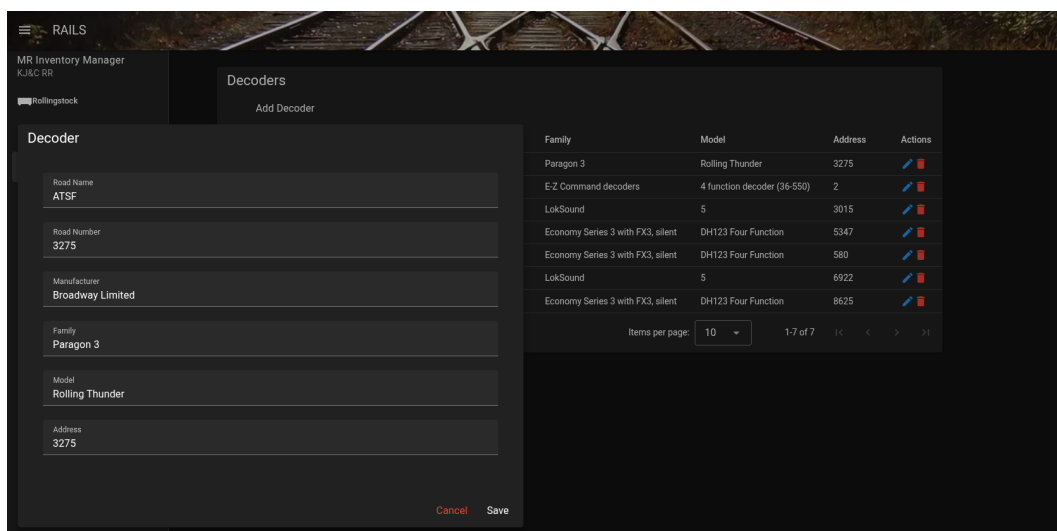


Figure 3.14: Edit Decoder

### 3.4.3 Deleting Decoder Records

To delete a decoder entry, click the red trash bin icon in the Actions column. A modal confirmation prompt appears requesting verification before the record is permanently removed from the database. Figure 3.15 illustrates the deletion confirmation dialog.

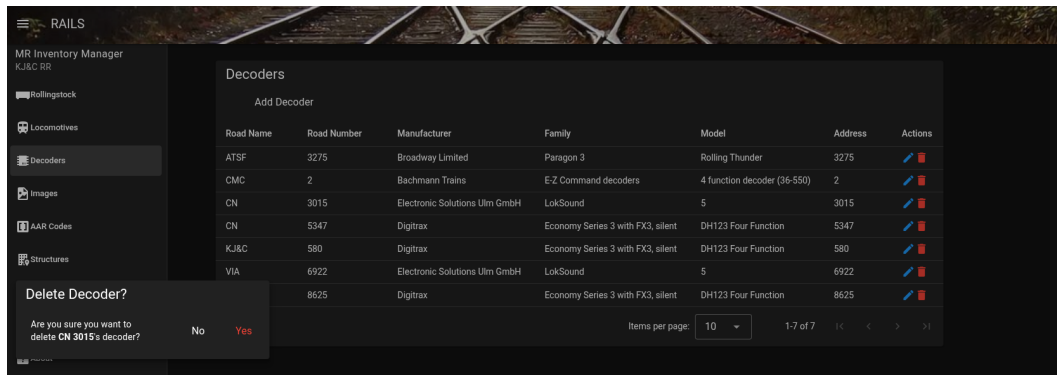


Figure 3.15: Delete Decoder Confirmation

## 3.5 Image Management

Selecting "Images" from the navigation drawer displays the Images management view. This page displays a tabular repository of all uploaded graphic assets stored in the Model Railroad File Manager (MRFM) microservice, detailing each item's Title, File Name, assigned Category (e.g., Rollingstock or Structure), and Action buttons. Figure 3.16 shows the main Images view.

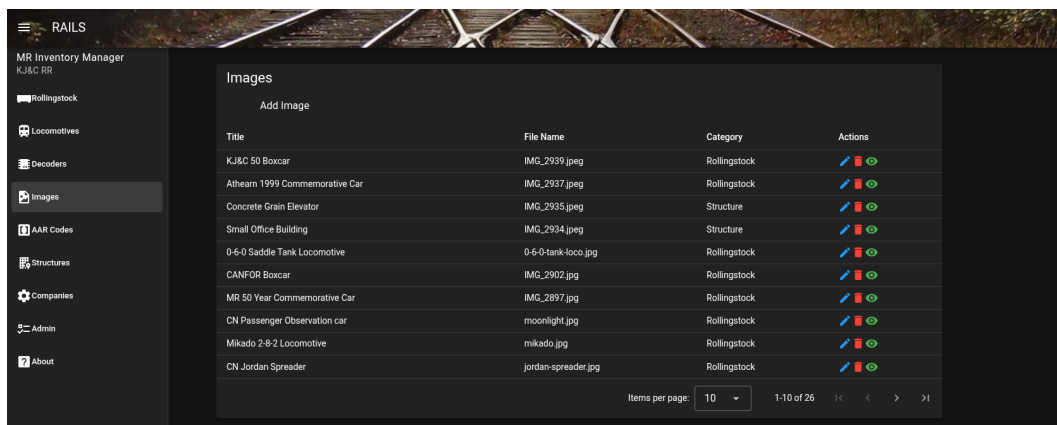


Figure 3.16: Images Page

### 3.5.1 Viewing Image Details

Clicking the green eye icon under the Actions column opens a modal dialog showing full meta-data and a visual preview of the selected asset. The dialog presents the Title, File Name,

Category, optional Notes, and an embedded image preview. Figure 3.17 shows the image viewer dialog.

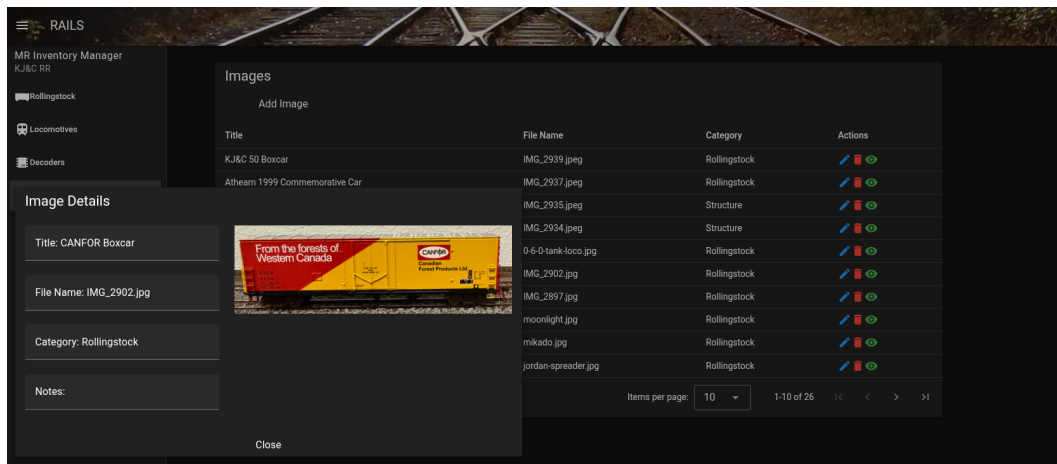


Figure 3.17: View Image Details

### 3.5.2 Adding a New Image

Clicking the “Add Image” button above the table opens a modal form for registering and uploading new graphic files/images. The form prompts for a Title, a local File input path, a Category assignment, and optional Notes. Figure 3.18 illustrates the upload modal.

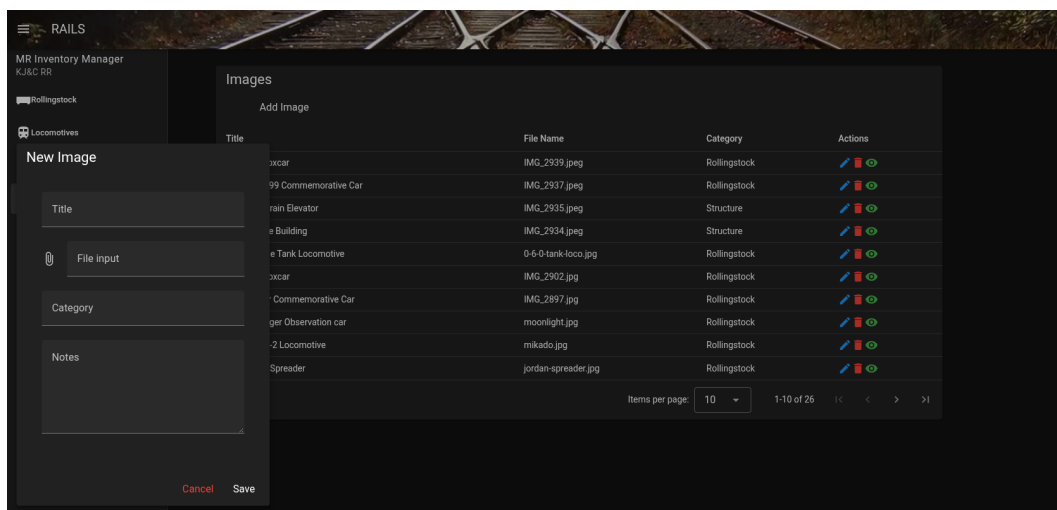


Figure 3.18: Add Image Dialog

### 3.5.3 Editing Image Metadata

To modify an existing image entry’s descriptive metadata, click the blue pencil icon in the Actions column/images. This opens the image dialog pre-populated with current values, allowing edits to the Title, File Name, Category, and Notes. Figure 3.19 shows the edit dialog.

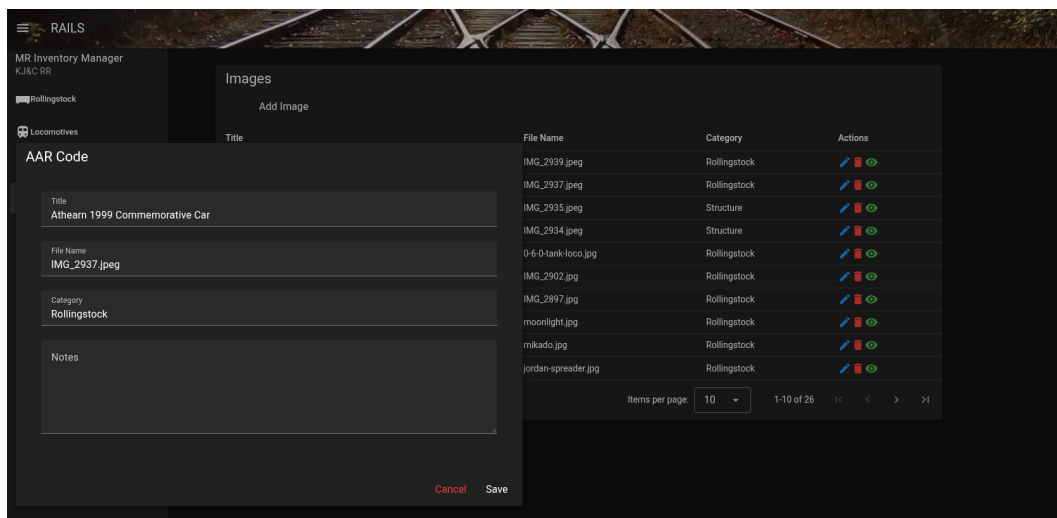


Figure 3.19: Edit Image Metadata

### 3.5.4 Deleting Image Records

To remove an image from the database repository, click the red trash bin icon under the Actions column in the images table. A modal confirmation prompt appears requesting verification before the image asset record is deleted. Figure 3.20 displays the deletion confirmation prompt.

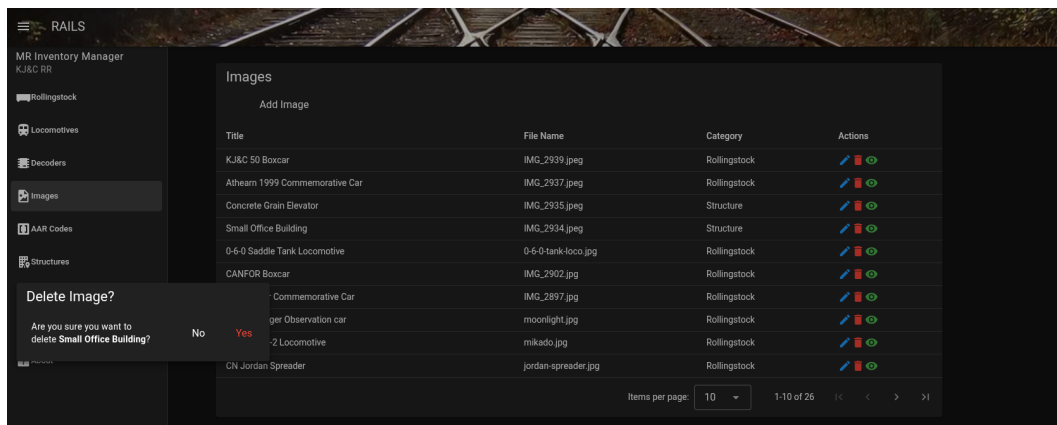


Figure 3.20: Delete Image Confirmation

## 3.6 AAR Code Reference

Selecting "AAR Codes" from the navigation drawer opens the Association of American Railroads (AAR) designation reference view. This page provides a standardized lookup table for rolling stock classification codes used throughout the inventory system, displaying the AAR Code, Rollingstock Type (RS Type), standard Description, and Action controls. Figure 3.21 illustrates the main AAR Codes reference table.

| AAR | RS Type       | Description                                                                                                                                                                                 | Actions |
|-----|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| BE  | Passenger     | A baggage car constructed and equipped to render it suitable for passenger train service, having side doors for the admittance of baggage or express, with or without windows or end doors. |         |
| EMR | Freight Car   | "Milk car that has ice bunkers"                                                                                                                                                             |         |
| DA  | Passenger Car | "Regular dining car, for the use of passengers in transit fitted with regular kitchen, tables, chairs or seats, with or without bar, carrying cooks and waiters."                           |         |
| DE  | Locomotive    | "Diesel electric locomotive"                                                                                                                                                                |         |
| FC  | Freight Car   | "A flat car, specifically equipped to carry trucks, trailers, containers, chassis, or removable trailer bodies for the transportation of freight in TOFC/COFC service."                     |         |
| FM  | Freight Car   | "Ordinary flat car for general service. This car has flooring laid over sills and without sides or ends."                                                                                   |         |
| GB  | Gondola       | "An open top car, having fixed sides, fixed or drop ends and solid bottom, suitable for mill trade. [includes former 'GK', 'GKA', 'GM' and 'GMA']"                                          |         |
| GS  | Gondola       | "An open top car, having fixed sides and ends and drop bottom, consisting of doors hinged at centre sills or side sills to dump outside and/or inside of rails."                            |         |
| GT  | Gondola       | "Open top car, having high fixed sides and fixed or hinged ends and solid bottom, suitable for unloading on dumping machines only."                                                         |         |
| HMA | Hopper        | "An Open Top Self-Clearing Car, having fixed sides and ends and bottom consisting of two divided hoppers with doors hinged lengthwise of car and dumping between rails."                    |         |

Figure 3.21: AAR Codes Page

### 3.6.1 Adding an AAR Code

Clicking the "Add AAR Code" button above the table opens a modal dialog to define a new equipment classification code. The form requires an AAR Code abbreviation, the associated RS Type, and a comprehensive Description detailing prototype car characteristics. Figure 3.22 shows the entry modal.

Figure 3.22: Add AAR Code Dialog

### 3.6.2 Editing AAR Codes

To modify an existing classification entry or update its prototype definition, click the blue pencil icon under the Actions column. This displays the AAR Code form with populated values for editing. Figure 3.23 shows the edit dialog interface.

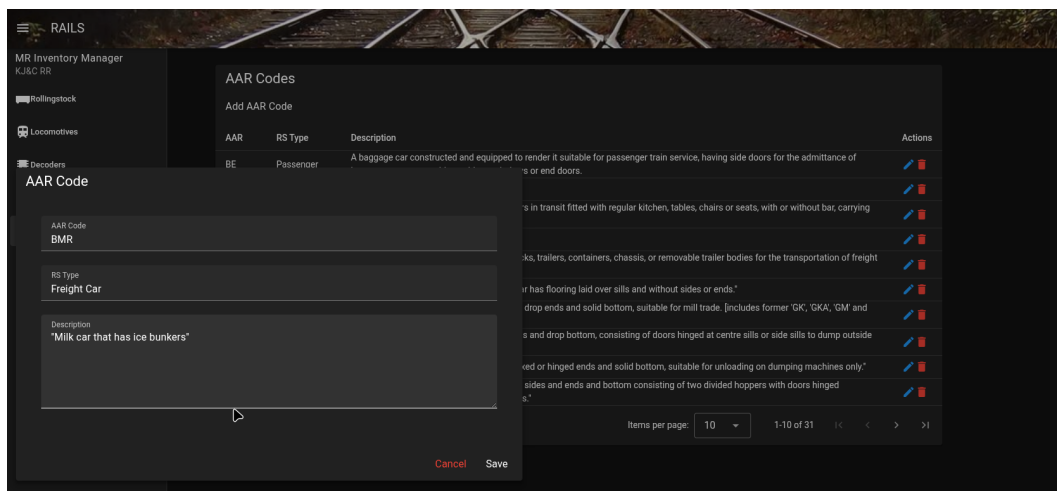


Figure 3.23: Edit AAR Code

### 3.6.3 Deleting AAR Codes

To remove a classification code from the system lookup table, click the red trash bin icon in the Actions column. A confirmation modal prompt appears requiring verification before the AAR code entry is permanently deleted. Figure 3.24 shows the deletion confirmation prompt.

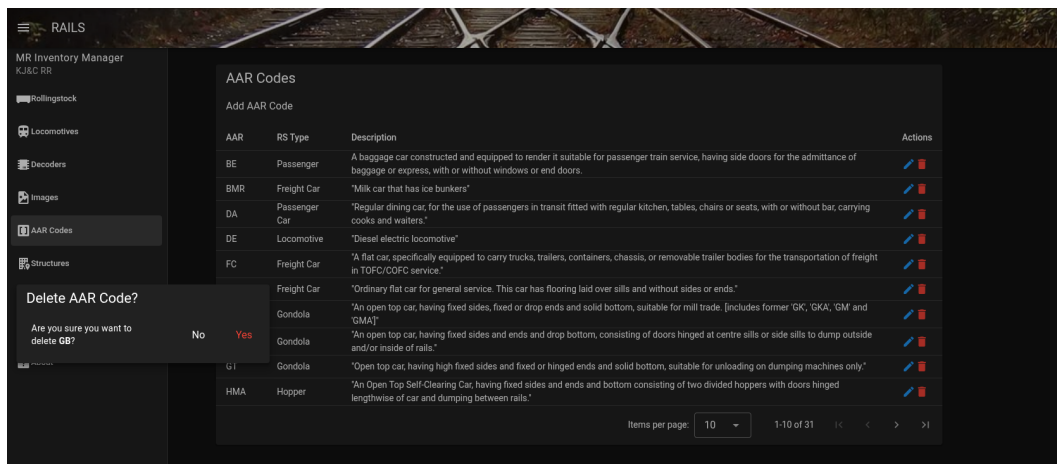


Figure 3.24: Delete AAR Code Confirmation

## 3.7 Structures Management

Selecting "Structures" from the navigation drawer displays the Structures management view. This page presents a tabular inventory of layout buildings and trackside structures, summarizing attributes such as Title, Use, Owner, layout Location, Year Built, and Action controls. Figure 3.25 shows the main Structures table page.

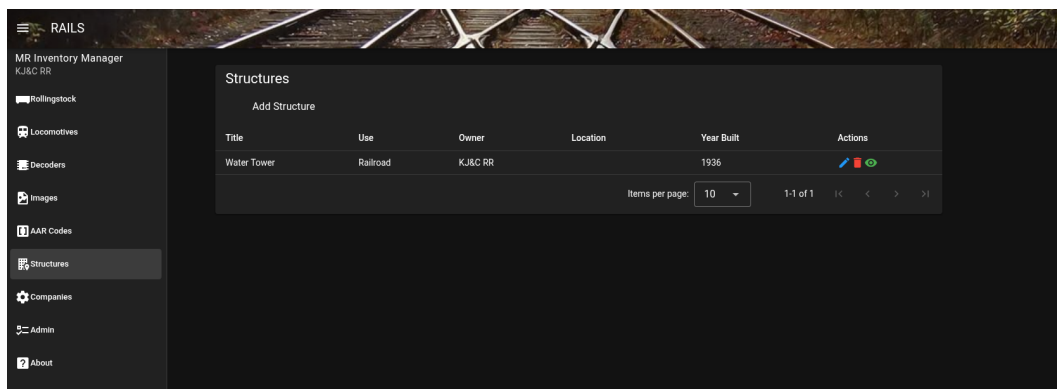


Figure 3.25: Structures Page

### 3.7.1 Viewing Structure Details

Clicking the green eye icon in the Actions column opens a detail view modal. This dialog presents recorded attributes alongside an embedded photo preview of the modeled structure. Figure 3.26 displays the structure inspection dialog.

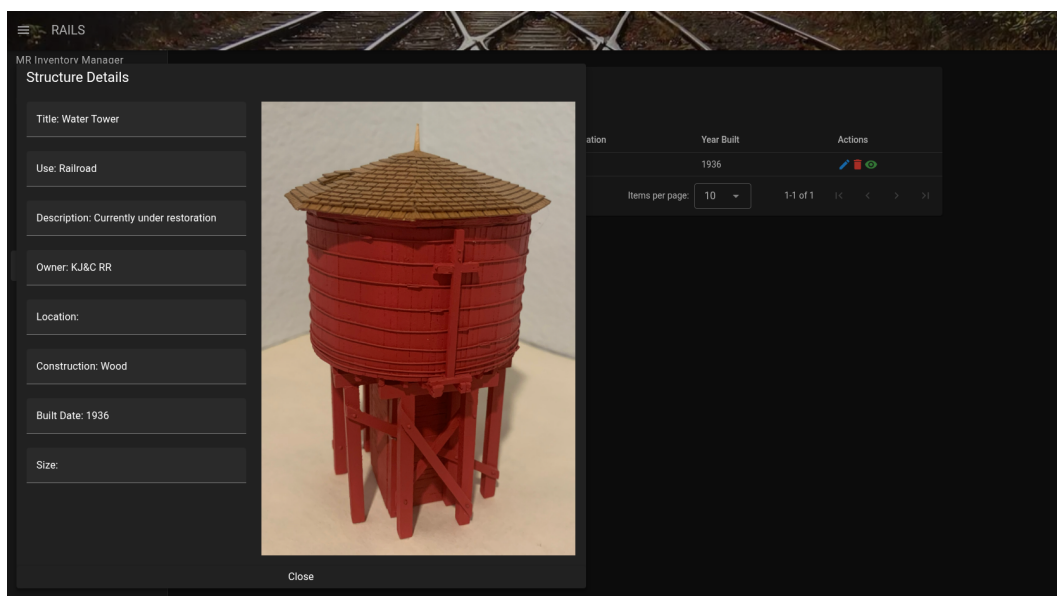


Figure 3.26: View Structure Details

### 3.7.2 Adding a New Structure

Clicking the "Add Structure" button above the table opens a modal dialog for adding a new building or facility to the database. The form includes comprehensive fields for general metadata (Title, Use, Owner, Location, Construction, Built Date, Size, Image) and operational parameters (Industrial classification, Type, Raw Materials, Capacities, Production Rates, Priority, AAR Codes In/Out, Load/Unload Durations, Siding Capacity, and Notes). Figure 3.27 illustrates the structure creation modal.

The image shows a 'New Structure' dialog box in the foreground, which is a dark-themed form with various input fields. The fields are arranged vertically and include: Title, Use, Description, Owner, Location, Construction, Built Date, Size, Image, Industrial (with a value of 'false'), Type, Raw Materials, rMCapacity, conRate, Priority, AAR Code In, Product, Product Capacity, Product Rate, AAR Code Out, Unload Duration, Load Duration, Siding Capacity, and a notes section at the bottom. At the bottom right of the dialog are 'Cancel' and 'Save' buttons. In the background, a table is visible with columns 'Location', 'Year Built', and 'Actions'. The 'Year Built' column contains the value '1936'. The 'Actions' column contains three icons: a blue pencil, a red square, and a green circle. Below the table, there is a pagination control showing 'Items per page: 10' and '1-1 of 1'.

| Location | Year Built | Actions                                                                                                                                                                                                                                                     |
|----------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | 1936       |    |

Items per page: 10 1-1 of 1 |< < > >|

Figure 3.27: Add Structure Dialog

### 3.7.3 Editing Structure Records

To update an existing entry, click the blue pencil icon under the Actions column. This opens the structure form populated with current records, allowing edits to physical attributes, image references, or industrial operational data. Figure 3.28 shows the edit structure dialog.

**Structure**

|                  |                             |               |          |                 |         |
|------------------|-----------------------------|---------------|----------|-----------------|---------|
| Title            | Water Tower                 | Use           | Railroad | Owner           | KJ&C RR |
| Description      | Currently under restoration |               | Location |                 |         |
| Construction     | Wood                        | Built Date    | 1936     | Size            |         |
| Image            | IMG_1430.jpg                | Industrial    | false    | Type            |         |
| Raw Materials    |                             | rMCapacity    |          | conRate         |         |
| Priority         |                             | AAR Code In   |          | Product         |         |
| Product Capacity |                             | Product Rate  |          | AAR Code Out    |         |
| Unload Duration  |                             | Load Duration |          | Siding Capacity |         |
| Notes            |                             |               |          |                 |         |

Cancel Save

Figure 3.28: Edit Structure

### 3.7.4 Deleting Structure Records

To remove a structure entry from the inventory, click the red trash bin icon in the Actions column. A confirmation prompt appears requiring verification before the record is deleted from the system database. Figure 3.29 illustrates the deletion modal dialog.

**RAILS**

MR Inventory Manager  
KJ&C RR

Rollingstock  
Locomotives  
Decoders  
Images  
AAR Codes  
Structures

**Structures**

Add Structure

| Title       | Use      | Owner   | Location | Year Built | Actions                                                                                                                                                                                                                                                           |
|-------------|----------|---------|----------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Water Tower | Railroad | KJ&C RR |          | 1936       |    |

Items per page: 10 1-1 of 1

**Delete Structure?**

Are you sure you want to delete Water Tower?

No Yes

Figure 3.29: Delete Structure Confirmation

## 3.8 Companies Reference

Selecting "Companies" from the navigation drawer opens the Companies management view. This page serves as a central reference table for railroad equipment manufacturers, locomotive builders, and operating railway companies. It displays the company short Name (abbreviation), Full Name,

Type classification (e.g., Builder, Railway), Location, and Action controls. Figure 3.30 illustrates the main Companies page interface.

| Name  | Full Name                           | Type    | Location     | Actions |
|-------|-------------------------------------|---------|--------------|---------|
| BBD   | Bombardier Inc.                     | Builder | Montreal, QC |         |
| CCF   | Canadian Car & Foundry              | Builder | Montreal, QC |         |
| CN    | Canadian National Railway           | Railway | Montreal, QC |         |
| CNR   | Canadian National Railway           | Railway | Montreal, QC |         |
| CP    | Canadian Pacific Railway            | Railway | Toronto, ON  |         |
| DOSCO | Dominion Steel and Coal Corporation | Builder | Halifax, NS  |         |
| ECC   | Eastern Car Company                 | Builder | Trenton, NS  |         |
| EMD   | Electro-Motive Diesel               | Builder | London, ON   |         |
| HST   | Hawker-Siddeley Transportation      | Builder | Trenton, NS  |         |
| MIL   | Marine Industries Limited           | Builder | Sorel, QC    |         |

Figure 3.30: Companies Page

### 3.8.1 Adding a Company

Clicking the “Add Company” button above the table opens a modal dialog to define a new corporate entry. The form prompts for the company’s short Name, Long Name, Type (such as Builder or Railway), and headquarters Location. Figure 3.31 shows the company entry modal.

| Name  | Full Name                           | Type    | Location     | Actions |
|-------|-------------------------------------|---------|--------------|---------|
| BBD   | Bombardier Inc.                     | Builder | Montreal, QC |         |
| CCF   | Canadian Car & Foundry              | Builder | Montreal, QC |         |
| CN    | Canadian National Railway           | Railway | Montreal, QC |         |
| CNR   | Canadian National Railway           | Railway | Montreal, QC |         |
| CP    | Canadian Pacific Railway            | Railway | Toronto, ON  |         |
| DOSCO | Dominion Steel and Coal Corporation | Builder | Halifax, NS  |         |
| ECC   | Eastern Car Company                 | Builder | Trenton, NS  |         |
| EMD   | Electro-Motive Diesel               | Builder | London, ON   |         |
| HST   | Hawker-Siddeley Transportation      | Builder | Trenton, NS  |         |
| MIL   | Marine Industries Limited           | Builder | Sorel, QC    |         |

Figure 3.31: Add Company Dialog

### 3.8.2 Editing Company Records

To modify an existing company entry, click the blue pencil icon under the Actions column. This opens the edit dialog populated with current information, allowing adjustments to reporting marks, full corporate titles, or location details. Figure 3.32 displays the edit company interface.

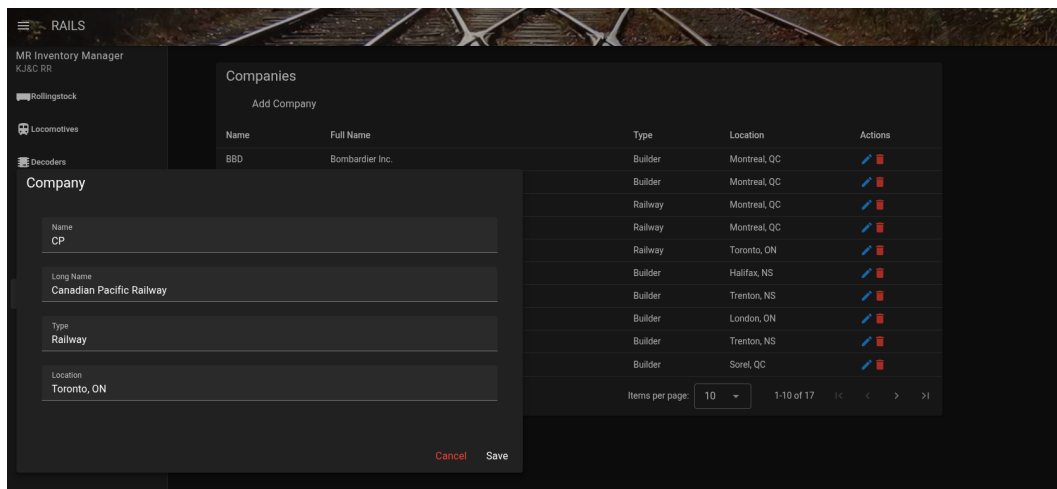


Figure 3.32: Edit Company

### 3.8.3 Deleting Company Records

To remove a corporate entity from the system reference table, click the red trash bin icon in the Actions column. A confirmation prompt appears asking for verification before removing the company entry. Figure 3.33 shows the deletion confirmation prompt dialog.

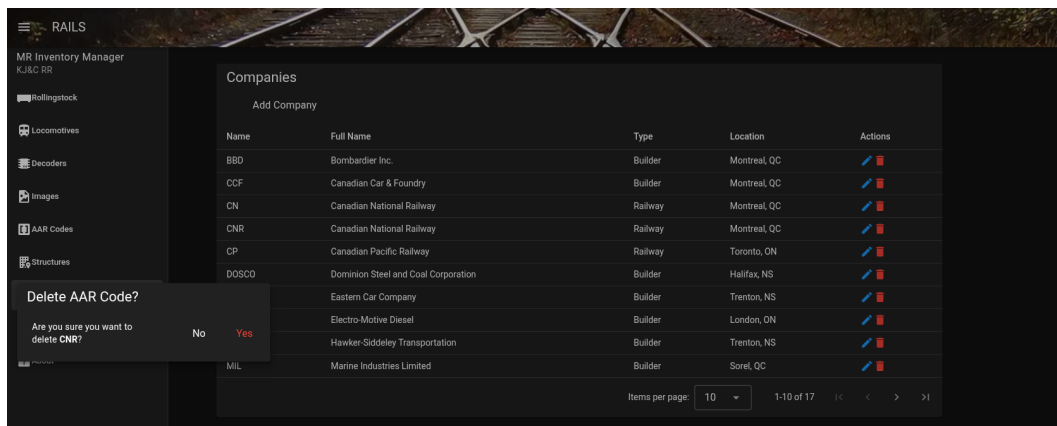


Figure 3.33: Delete Company Confirmation

## 3.9 Admin Dashboard

Selecting "Admin" from the navigation drawer opens the Admin dashboard view. This page provides centralized controls for system data management, including database collection exports, imports, report generation, and car card printing. Figure 3.34 shows the main Admin page interface.

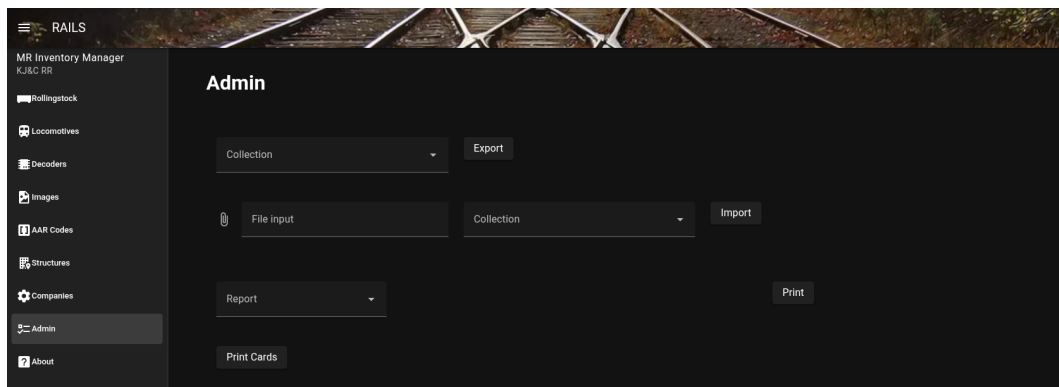


Figure 3.34: Admin Page

### 3.9.1 Exporting Data Collections

The Export control allows backup or data extraction of individual system collections. Selecting the "Collection" dropdown opens a list of available data sets, including AAR Codes, Companies, Images, Rolling Stock, Decoders, and Structures. Figure 3.35 illustrates the collection selection menu for data export.

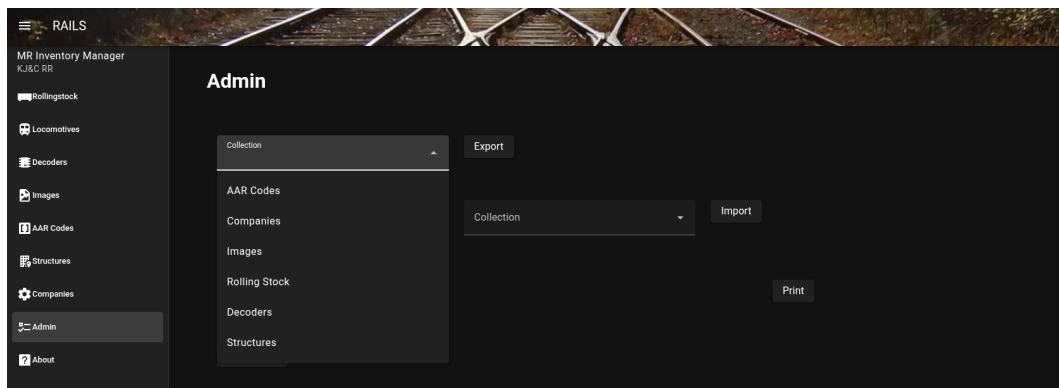


Figure 3.35: Select Export Collection

### 3.9.2 Importing Data Collections

To restore or bulk-load records, click the paperclip icon or file input field to launch the system file browser and select a local source file. Figure 3.36 shows the native file chooser selection dialog.

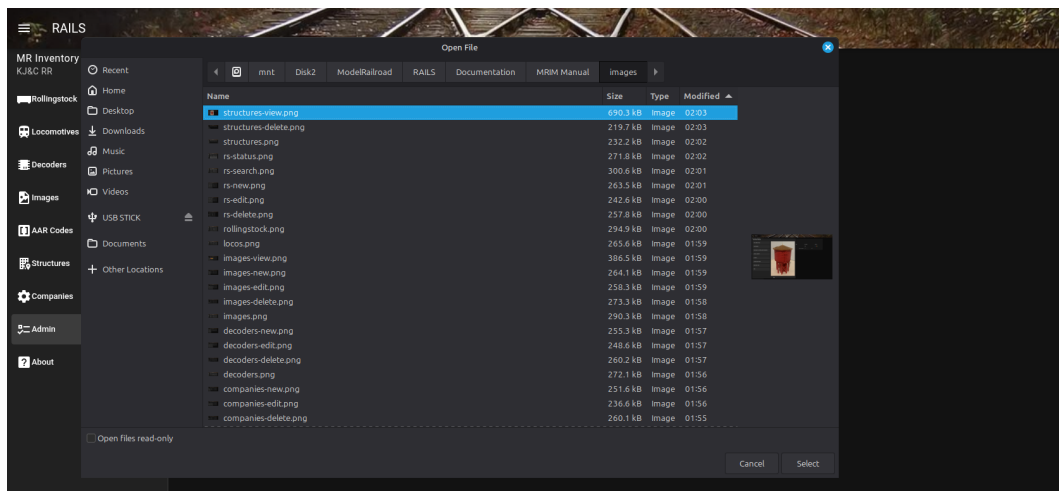


Figure 3.36: Select Import Source File

After specifying a source file, select the target database destination using the “Collection” dropdown menu. Options include AAR Codes, Companies, Images, Rolling Stock, Decoders, and Structures. Figure 3.37 illustrates selecting the target import collection before clicking “Import”.

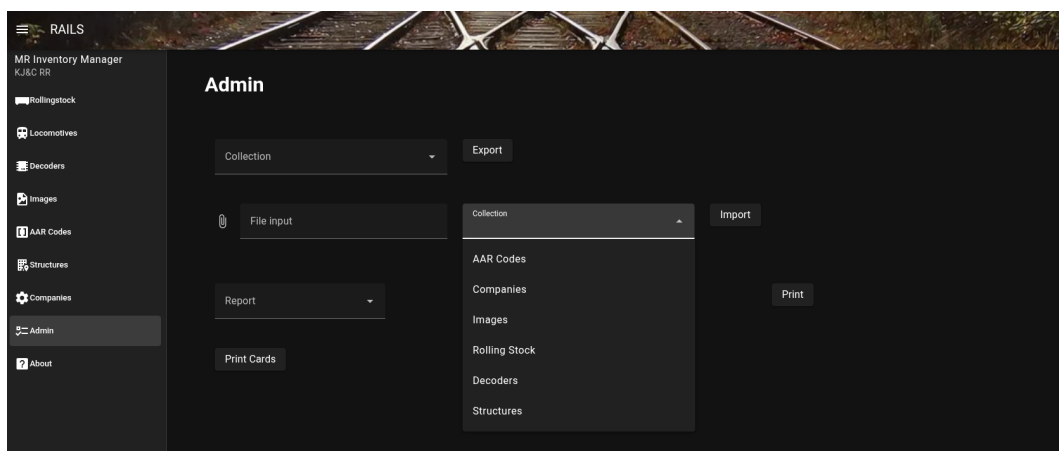


Figure 3.37: Select Target Import Collection

### 3.9.3 Generating System Reports

The Report section generates formatted system documentation and operational materials as a Portable Document Format (PDF) format. Selecting the “Report” dropdown displays available system inventories and assets for export, such as Rolling Stock, AAR Codes, Companies, Images, RFID, and Structures. Figure 3.38 shows the report type selection menu.

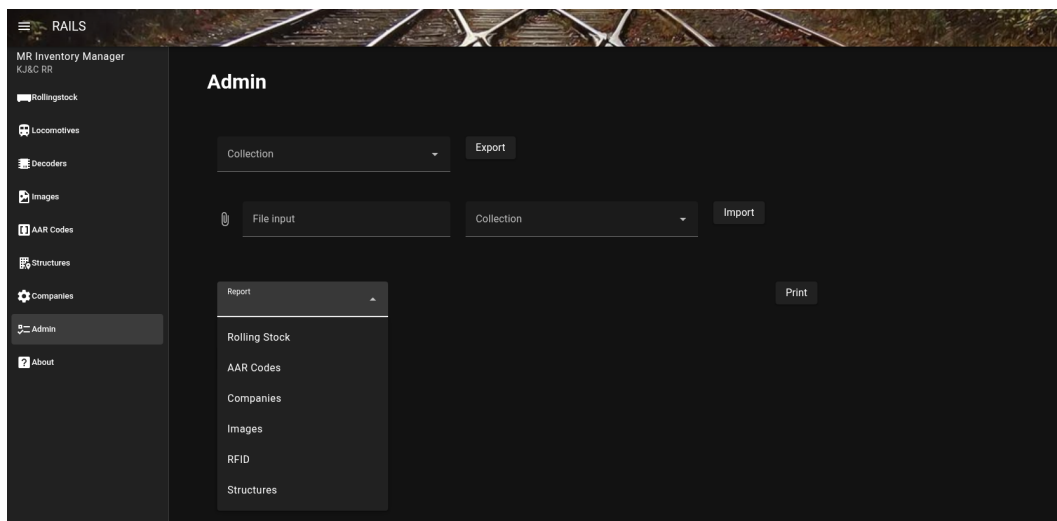


Figure 3.38: Select Report Category

When a specific report category like Rolling Stock is selected, additional sorting options appear. Clicking the "Sorted By" dropdown allows records to be ordered by parameters such as Road Names, AAR Codes, or Status. Figure 3.39 illustrates the sorting criteria options.

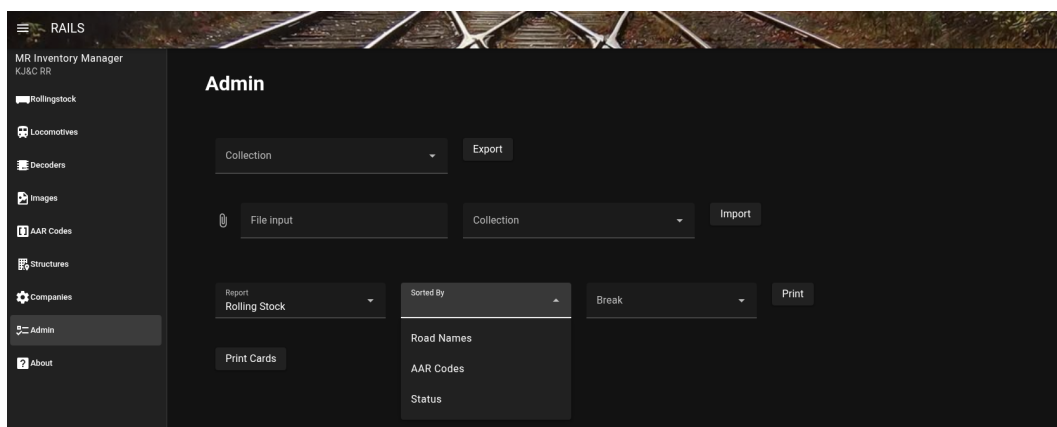


Figure 3.39: Report Sort Options

Reports can also be structured using page breaks or formatting style options. Selecting the "Break" dropdown provides options for Continuous formatting, Table views, or explicit Page breaks between sections. Figure 3.40 shows the report page break selection menu.

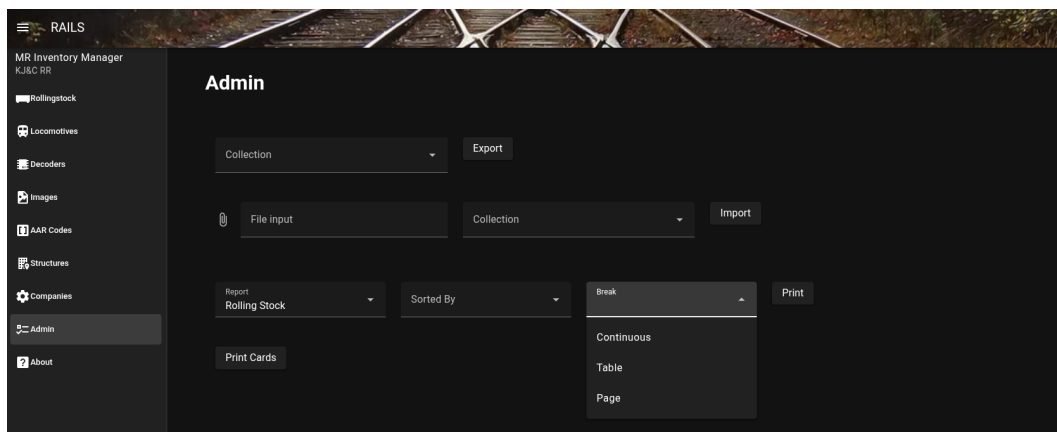


Figure 3.40: Report Page Break Formatting

Clicking on the "Print Cards" creates a set of pages that can be cut into car cards used as a two-part system to track freight cars and simulate real-world cargo movements across a layout.

## Appendix A

# Railroad Inventory

### A.1 aarcodes

An Association of American Railroads (AAR) code refers to a classification code used by the AAR to identify different types of equipment and rollingstock used in the North American railway industry. The AAR codes are a standardized way of categorizing and labeling various railway vehicles, making it easier to identify and track equipment across different railways and organizations.

Table A.1: AarCode Collection Fields Table

| Field            | Type   | Description                                                                                                                                     | Req'd |
|------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| aarCode          | String | A classification code used by the AAR to identify different types of rollingstock                                                               | Yes   |
| rollingstockType | String | A category of rollingstock having common characteristics, such as; Auxiliary, Freight, Gondola, Hopper, Locomotive, Non Revenue, and Passenger. | No    |
| description      | String | A written representation that aims to make vivid the characteristics of the AAR Code.                                                           | No    |

### A.2 dccs

Digital Command and Control (DCC) is a system used in model railroading to control and operate model trains and their accessories. It is a standardized method of transmitting control signals digitally to locomotives and other rollingstock. DCC allows for more precise and independent control of multiple trains on the same track.

Table A.2: Dccs Collection Fields Table

| Field        | Type      | Description                                                                                                                 | Req'd |
|--------------|-----------|-----------------------------------------------------------------------------------------------------------------------------|-------|
| locomotiveID | ObjecctId | The identity object of the dccs document assigned by MongoDB at the time the rollingstock (locomotive) document is created. | Yes   |

*Continued on next page*

Table A.2 – Continued from previous page

| Field   | Type   | Description                                                                                                                                                                                     | Req'd |
|---------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| mfg     | String | The company that made the DCC decoder installed in a rollingstock, typically a locomotive.                                                                                                      | No    |
| family  | String | A group of DCC decoders produced by a specific manufacturer that share common features, specifications, and compatibility.                                                                      | No    |
| model   | String | A DCC decoder offering specific features or specifications within the same product family                                                                                                       | No    |
| address | String | A unique identifier assigned to each DCC decoder, which is a numerical or digital label that distinguishes one rollingstock from another in the DCC system. DCC addresses range from 1 to 9999. | No    |

### A.3 images

A set of photographs or illustrations of the rollingstock and structures of the model railroad present in RAILS. These images may capture different designs, configurations, and features of rollingstock, providing a visual representation of the diversity within the railway transportation system.

Table A.3: Images Collection Fields Table

| Field    | Type   | Description                                                                                                       | Req'd |
|----------|--------|-------------------------------------------------------------------------------------------------------------------|-------|
| fileName | String | A unique identifier assigned to a computer file in order to distinguish it from other files within a file system. | Yes   |
| title    | String | The name or label given to the image to identify and distinguish it from others.                                  | No    |
| notes    | String | A brief record of events or observations about the image.                                                         | No    |
| category | String | A classification or group of images that share common characteristics, features, or attributes.                   | No    |

### A.4 industries

Sectors of economic activity that involve the production of goods or the provision of services. These sectors are characterized by similar business activities and are often grouped together based on their common features. Railways service a wide range of industries, facilitating the transportation of goods and passengers. The industries collection is used to identify the industries that are present in the model railroad system.

Table A.4: Industries Collection Fields Table

| Field            | Type   | Description                                                                                                                                         | Req'd |
|------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| shortName        | String | A concise or abbreviated version of a longer name. It could be a nickname, an acronym, or a shortened form that is used for convenience or brevity. | Yes   |
| longName         | String | A word or set of words by which an industry is identified and distinguished from others, legal name.                                                | No    |
| industryType     | String | The various categories or sectors of economic activity in which businesses and organizations engage.                                                | No    |
| industryLocation | String | A specific place or position in space, often designated by its geographical coordinates or described in relation to other landmarks.                | No    |

## A.5 rollingstocks

rollingstock refers to the vehicles that move on a railway track. These vehicles are an essential part of any railway system and include various types used for transporting goods and passengers. Rollingstock can be broadly categorized into two main types: passenger rollingstock and freight rollingstock. Locomotives are considered a type of rollingstock in the context of railway transportation.

Table A.5: Rollingstock Collection Fields Table

| Field       | Type   | Description                                                                                                                                                                                                   | Req'd     |
|-------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| roadName    | String | The road name also referred to as reporting marks are a set of letters used to identify the owner or operator of a railcar or locomotive.                                                                     | Yes       |
| roadNumber  | String | The road number is an identifier for a railcar or a locomotive and is numerical.                                                                                                                              | Yes       |
| color       | String | The exterior paint or livery applied to the rollingstock. The color of rollingstock serves several purposes, including identification, branding, and visibility.                                              | No        |
| aarCode     | String | See paragraph A.1                                                                                                                                                                                             | No        |
| description | String | The informative account that provides information about the characteristics, features, qualities, or attributes of the rollingstock.                                                                          | No        |
| numberBlt   | Long   | The total quantity or count of a particular rollingstock.                                                                                                                                                     | No<br>(0) |
| inSvcDate   | Date   | The specific date on which a piece of rollingstock is officially put into operation or use. It marks the beginning of its active service life, during which it is expected to perform its intended functions. | No        |

*Continued on next page*

Table A.5 – Continued from previous page

| Field                                                                        | Type   | Description                                                                                                                                                                                                                                                                                                                                                                                                          | Req'd     |
|------------------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| insideLength                                                                 | String | For passenger cars it is measured from one end of the passenger compartment to the other. It defines the available space for seating and other passenger-related amenities. For freight cars it is measured from the inside faces of the end walls, excluding any bulkheads or projections that might reduce the usable length. It provides an indication of the space available for loading and transporting goods. | No        |
| insideHeight                                                                 | String |                                                                                                                                                                                                                                                                                                                                                                                                                      | No        |
| insideWidth                                                                  | String |                                                                                                                                                                                                                                                                                                                                                                                                                      | No        |
| loadTypes                                                                    | String |                                                                                                                                                                                                                                                                                                                                                                                                                      | No        |
| capacity                                                                     | Long   | Capacity refers to the maximum amount of cargo, typically measured in weight or volume, that it can safely and efficiently transport.                                                                                                                                                                                                                                                                                | No<br>(0) |
| bldr                                                                         | String |                                                                                                                                                                                                                                                                                                                                                                                                                      | No        |
| bltDate                                                                      | Date   |                                                                                                                                                                                                                                                                                                                                                                                                                      | No        |
| notes                                                                        | String | A brief record of events or observations about the rollingstock.                                                                                                                                                                                                                                                                                                                                                     | No        |
| ltWeight                                                                     | Long   | The weight of the car itself without any cargo or load.                                                                                                                                                                                                                                                                                                                                                              | No<br>(0) |
| loadLimit                                                                    | Long   | The maximum weight that a freight car can carry, including both the weight of the car itself (tare weight or lightweight) and the weight of the cargo (payload).                                                                                                                                                                                                                                                     | No<br>(0) |
| lastMaintDate                                                                | Date   |                                                                                                                                                                                                                                                                                                                                                                                                                      | No        |
| locationNow                                                                  | String |                                                                                                                                                                                                                                                                                                                                                                                                                      | No        |
| homeLocation                                                                 | String |                                                                                                                                                                                                                                                                                                                                                                                                                      | No        |
| rsStatus                                                                     | String | The current condition, availability, or operational state of the rollingstock used on a railway.                                                                                                                                                                                                                                                                                                                     | No        |
| numAxles                                                                     | Long   | The total count of transverse metal rods or central shafts that span beneath the rollingstock to support its weight and connect opposite wheel assemblies.                                                                                                                                                                                                                                                           | No<br>(4) |
| imageID                                                                      | String | The file name of the image; See paragraph A.3                                                                                                                                                                                                                                                                                                                                                                        | No        |
| modelWeight                                                                  | Double | The mass of the model rollingstock in either grams or ounces.                                                                                                                                                                                                                                                                                                                                                        | No<br>(0) |
| modelLength                                                                  | Double | The distance measure between the middle of the front coupler knuckle to the middle of the rear coupler knuckle of the model rollingstock.                                                                                                                                                                                                                                                                            | No<br>(0) |
| rfid                                                                         | String | The 10 character identifier emitted by the RFID tag attached to the model rollingstock.                                                                                                                                                                                                                                                                                                                              | No        |
| rfidLocation                                                                 | Double | The distance the RFID tag is attached to the rollingstock measured from the A end.                                                                                                                                                                                                                                                                                                                                   | No<br>(0) |
| Note: The numbers in parentheses in the Req'd column are the default values. |        |                                                                                                                                                                                                                                                                                                                                                                                                                      |           |

## A.6 structures

| Field          | Type    | Description | Req'd |
|----------------|---------|-------------|-------|
| title          | String  |             | Yes   |
| structureUse   | String  |             | Yes   |
| description    | String  |             | No    |
| owner          | String  |             | No    |
| location       | String  |             | No    |
| construction   | String  |             | No    |
| builtDate      | String  |             | No    |
| size           | String  |             | No    |
| image          | String  |             | No    |
| isIndustrial   | Boolean |             | No    |
| rawMaterials   | String  |             | No    |
| rMCapacity     | String  |             | No    |
| conRate        | String  |             | No    |
| priority       | String  |             | No    |
| aarCodeIn      | String  |             | No    |
| product        | String  |             | No    |
| productCap     | String  |             | No    |
| prodRate       | String  |             | No    |
| aarCodeOut     | String  |             | No    |
| unloadDuration | String  |             | No    |
| loadDuration   | String  |             | No    |
| sidingCap      | String  |             | No    |
| notes          | String  |             | No    |

Table A.6: Structures Collection Fields Table

# Acronyms

**Association of American Railroads (AAR)** 32

**Digital Command and Control (DCC)** 32, 33

**Graphic User Interface (GUI)** is the visual way you interact with a computer program using elements like windows, icons, menus, and buttons, instead of typing commands. GUIs are the standard way most people interact with computers today 6

**Model Projects and Purchase Manager (MPPM)** 3, 8

**Message Queuing Telemetry Transport (MQTT)** is a lightweight messaging protocol designed for machine-to-machine (M2M) communication in resource-constrained environments, like those found in the Internet of Things (IoT) 6

**Model Railroad (MR)** 4

**Model Railroad File Manager (MRFM)** 4

**Model Railroad Inventory Manager (MRIM)** 3, 4, 5, 8

**Model Railroad Layout Manager (MRLM)** 3, 8

**Portable Document Format (PDF)** 29

**Railway Administration and Information Logical System (RAILS)** 3, 4, 6, 7, 8, 33

**Representational State Transfer (REST)** is a set of architectural principles for designing web services. It's not a protocol itself (like HTTP), but rather a guideline for creating web services 4

**Radio Frequency Identification (RFID)** is technology that uses radio waves to wirelessly identify and track objects 9

**Railroad Inventory Data Services (RIDS)** 4

**rollingstock (RS)** all the vehicles that operate on a railroad line. This includes both powered and unpowered vehicles 4

**Rollingstock RFID Manager (RSRM)** 3, 8, 9

**Single Page Application (SPA)** a web application that loads a single HTML page in the user's browser and dynamically updates the content based on user interaction, without reloading the entire page. This creates a more fluid and responsive user experience, similar to what you might expect from a native mobile app 3, 4, 5, 6, 7, 8

**Transmission Control Protocol/Internet Protocol (TCP/IP)** a suite of communication protocols that governs how data is exchanged over the internet or a private network. It is the foundation of any data exchange on the Web 5

**yet another markup language (YAML)** is a human-readable data serialization language used for configuration files, data exchange, and system automation, known for its use of indentation to define structure, similar to Python, and for being a strict superset of JSON 6, 8